KERAAL Team Meeting Agenda

Last team session before the midterm presentation · Fri 2026-10-16 · 90 min · Lead · Uijin · Roshini · Joshi

Three things we must finish today. ① Collect the 10/16 results and decide what goes into the presentation ② Agree on one research question (the story of the talk) ③ Assign slide owners and draft deadlines.
Rule. One slide (one table) per person for results. Unfinished is fine: just say how far you got. Debugging happens after the meeting.

1. Schedule

Time	Item	How	Outcome
0–5	Presentation format	Lead: date, length, format, what to submit	Talk length, number of slides
5–30	10/16 results	Roshini 10 min → Uijin 7 → Joshi 5 → questions 3	For each item: "in the talk / not"
30–40	New findings	Lead: 2 figures + the four points in section 3	Wording to fix, use of B codes
40–55	**Research question**	Lead's proposal → objections and edits → decision (do not shorten)	One-sentence story everyone agrees on
55–75	Slide assignment	Put the table in section 5 on screen and fill in names	Owners, draft deadlines
75–90	Buffer · next week	Absorbs overrun. If time remains: schedule 10/23 items (U3, U4, R3, R4)	Next week's tasks

2. What to bring

Who	Bring	Check (from the Team Roles document)
Roshini	**R1** threshold calibration results table **R2** P1 raw-advantage check table	R1: does the pre-calibration score match the confirmatory result; does the calibrated score beat the 95% participant-permutation null? R2: do static features alone (body size, camera position) reproduce the R0 score (0.737)?
Uijin	**U1** cleaned data v2 **U2** time-segment table `temporal_segments.csv`	U1: Kinect values match the originals; 2A S1 split by site U2: all 98 Temporal-error segments; frame = annotation time × 15; ranges inside recording length
Joshi	10 screenshots (4 CTK + 6 ELK), answers	Is CTK E3 a different movement from E1? What is different about ELK P2? ("I don't know" is a valid answer.)
Lead	Presentation format, 2 figures	`CTK_acted_vs_real.png`, `G3_CTK_E1_B1-B3.png` in `explore/video_check/`

3. New findings (video check 10/13, exploratory)

Finding	Details	Source
2A are healthy adults	1A = patients, 2A = healthy adults (both robot-coached), 3 = healthy people acting out errors. Fix anything that says "1A + 2A = patients".	Official readme (`group3/readme_files_format.txt`)
G3 B code = body part	B1 left arm, B2 right arm, B3 both arms (15/15 in CTK and RTK E1). This partly changes "G3 has no body-region labels" in our homework → body-region experiments (3 and 4) become partly possible. Not yet checked for E2, E3, ELK.	`video_check/README.md` §3
Acted vs real	CTK E1 wrist height: acted correct 0.78 · acted error 0.19 · patient correct 0.49 · patient error 0.37 · annotators disagree 0.42. Patients' correct movements fall below the boundary learned from acted data → why 92–97% were called errors. *Limits: metric chosen after seeing results; 1A (Brest) stand, G3 sit.*	`video_check/README.md` §4, `CTK_acted_vs_real.png`
fps wording	Frame = annotation time × 15 is certain. True capture rate (10 vs 15 fps) is not confirmed → label any seconds as "annotation time".	`video_check/README.md` §1

4. Decision: the research question of the talk

	Details
Original question (homework)	"Do normalized kinematic features improve subject-independent error classification compared with raw coordinates?" Train on G3, test on 1A and 2A.
Results so far	Pre-registered H1–H3 not supported (H1 Macro-F1 0.407, 95% permutation null 0.428). The model called 92–97% of test recordings errors. The reverse direction (real → acted) gives AUC 0.71–0.77. Annotators A and B differ by about 12° in their RTK threshold.
Lead's proposal	**"A model trained on acted data failed not because it cannot recognize errors, but because the decision threshold differs."** Flow: original question → pre-registered test (failed) → cause (threshold) → being tested with calibration (R1). We show the failure openly and present pre-registration as a strength.
Depending on R1	Recovers after calibration → "threshold difference is the main cause". Weak recovery → "threshold hypothesis still being tested".
Decide today	Do we agree? Any wording to change? Which results do we label "exploratory" in the talk?

5. Slide plan (assuming a 15-minute talk; adjust to the format)

#	Content	Source	Owner	Draft due
1	Problem: home rehab, nobody tells you when you do it wrong	Project Overview §1		

#	Content	Source	Owner	Draft due
2	Data: group table (2A = healthy), 3 exercises, error definitions	Official readme, homework §2		
3	Question and pre-registered design (train G3 → test 1A, 2A)	PREREGISTRATION.md		
4	Result 1: confirmatory test failed (92–97% called errors)	PREREGISTRATION.md Results, PROGRESS 10/08		
5	Result 2: reverse direction works (AUC 0.71–0.77)	KERAAL_Kinect_Model_Report.pdf		
6	Cause: threshold differences (acted vs real figure, annotators 12°, Joshi's screenshots)	video_check, Team Roles R3, Joshi's task		
7	In progress: threshold calibration (R1), time-segment model (R4)	Today's results		
8	Next steps and limitations (exploratory labels, small sample, posture/camera differences)	PROGRESS.md backlog		

6. Decision log (fill in during the meeting)

Decision	Owner	Due	Notes
Research-question sentence			
Include R1 · R2 results?			
Use U1 · U2?			
Slide(s) for Joshi's screenshots			

Common rules: label every post-confirmatory result "exploratory" · split evaluation by participant · never change `configs/p1_confirmatory.json` · videos and screenshots are for the class presentation only, never uploaded (CC BY-NC-SA). Record the meeting outcome as one paragraph in PROGRESS.md.